

UNCERTAINTY ESTIMATION AND CONFIDENCE CALIBRATION IN UNIMOD FOR MULTIMODAL MEDICAL DIAGNOSIS

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UNCERTAINTY ESTIMATION AND CONFIDENCE CALIBRATION IN UNIMOD FOR MULTIMODAL MEDICAL DIAGNOSIS

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Abstract

Multimodal learning has become an important approach in medical artificial intelligence. Enabling diagnostic systems to combine information from different sources such as medical images and clinical text. Although these systems can improve diagnostic performance and their predictions may be unreliable when the input data are incomplete, ambiguous or inconsistent. In medical applications, overconfident predictions can be particularly risky because they may influence clinical decision making. Therefore, estimating uncertainty and ensuring that a model's confidence accurately reflects its likelihood of being correct are essential for building more reliable medical AI systems. This thesis aims to investigate uncertainty estimation and confidence calibration in UniMod, which is a multimodal medical diagnosis framework. The study will examine whether the model's confidence scores correspond to its actual diagnostic accuracy and whether calibration techniques can improve the reliability of its predictions. A baseline UniMod model will be evaluated using suitable medical image text data and subject to dataset availability. Calibration methods such as temperature scaling where applicable will be explored to reduce the gap between predicted confidence and observed correctness. In addition, selective prediction will be investigated to assess whether the model can identify uncertain cases and defer them for further review rather than making potentially unreliable predictions. The proposed system will be evaluated using diagnostic performance metrics alongside calibration measures such as Expected Calibration Error and the Brier score. Selective prediction performance will also be assessed by examining the relationship between prediction coverage and error rate. The results are expected to provide insight into the reliability of UniMod's predictions and the effectiveness of calibration techniques in multimodal medical diagnosis. This research aims to contribute to the development of more trustworthy medical AI systems by emphasizing not only diagnostic accuracy but also the reliability of model confidence.

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We hereby certify that **Raian Mahadi**, **Mehrab Hasan Badhon**, **Mamun Shake** students of the Department of Computer Science and Engineering (CSE) at Uttara University, have successfully completed their thesis titled "**Uncertainty Estimation and Confidence Calibration in UniMod for Multimodal Medical Diagnosis**" in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering (CSE) at Uttara University.

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We, the undersigned, hereby declare that this project entitled "**Uncertainty Estimation and Confidence Calibration in UniMod for Multimodal Medical Diagnosis**" presents the original thesis work of Raian Mahadi, Mehrab Hasan Badhon and Mamun Shake. Wherever the contributions of others are involved, due acknowledgment has been made through proper citations and references. This work was carried out under the supervision of Mohsin Hossain, in the Department of Computer Science and Engineering (CSE) at Uttara University.

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Dedication

To our parents and our family. Both our parents give enough inspiration and encouragement to complete our thesis work.

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Symbols

x	denotes	the input medical image;
t	denotes	the clinical text input;
y	denotes	the true diagnostic label;
$\hat{y}$	denotes	the predicted diagnostic label;
p	denotes	the predicted probability;
$\hat{p}$	denotes	the calibrated predicted probability;
f_θ	denotes	the model parameterized by θ;
θ	denotes	the model parameters;
z	denotes	the logit vector produced by the model;
T	denotes	the temperature scaling parameter;
C	denotes	the prediction confidence;
U	denotes	the predictive uncertainty;
N	denotes	the total number of evaluated samples;
K	denotes	the number of diagnostic classes;
n_k	denotes	the number of samples in calibration bin k;
B	denotes	the number of confidence calibration bins;
ECE	denotes	the Expected Calibration Error;
BS	denotes	the Brier Score;
Acc	denotes	the classification accuracy;
R	denotes	the selective risk or error rate;
κ	denotes	the prediction coverage rate;
$\mathbb{E}$	denotes	the expected value;
$\log$	denotes	the natural logarithm;
$\arg\max$	denotes	the argument that maximizes a function;
σ	denotes	the softmax probability function;
τ	denotes	the confidence threshold for accepting a prediction;
$\mathcal{D}$	denotes	the dataset;
$\mathcal{L}$	denotes	the loss function.

List of Abbreviation

AI	Artificial Intelligence;
ML	Machine Learning;
DL	Deep Learning;
UniMod	Unified Multimodal Medical Diagnosis Framework;
MM	Multimodal;
MedAI	Medical Artificial Intelligence;
ECE	Expected Calibration Error;
BS	Brier Score;
Acc	Accuracy;
NLL	Negative Log Likelihood;
OOD	Out of Distribution;
UQ	Uncertainty Quantification;
MLP	Multilayer Perceptron;
CNN	Convolutional Neural Network;
NLP	Natural Language Processing;
AUROC	Area Under the Receiver Operating Characteristic Curve;
AUPRC	Area Under the Precision-Recall Curve;
TP	True Positive;
TN	True Negative;
FP	False Positive;
FN	False Negative.

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Chapter 1:

Introduction

1.1 Background of the Study

Purpose: Introducing the research topic and explain why it matters.

Content: Artificial Intelligence (AI) is increasingly being used in healthcare to help analyze medical images and support disease diagnosis. Some systems use both medical images and clinical text to make predictions. UniMod is a multimodal medical diagnosis framework designed to learn from these different sources of information. However, making an accurate prediction is not the only challenge. A model may sometimes be highly confident even when its prediction is wrong. Confidence calibration helps determine whether a model's confidence matches its actual performance. While uncertainty estimation helps identify predictions that may be less reliable. This study will investigate these issues in UniMod to better understand the reliability of its diagnostic predictions.

Example:

A model predicts a disease with 90% confidence but its predictions at that confidence level are correct only 70% of the time. This may indicate that the model is overconfident. [**?**].

1.2 Problem Statement

Purpose: Explaining the main problem the research will address.

Content: Although multimodal models can support medical diagnosis. Their confidence scores may not always reflect how accurate their predictions are. This can make it difficult to identify potentially unreliable results. This study will evaluate the confidence calibration of UniMod and investigate whether a suitable calibration technique such as temperature scaling can improve the reliability of its confidence scores.

Example:

If UniMod assigns high confidence to incorrect predictions, calibration techniques may help make its confidence scores more representative of its actual performance.[**?**]

1.3 Research Objectives

Purpose: State what the research aims to achieve

Content: The main objective is to investigate uncertainty estimation and confidence calibration in UniMod for multimodal medical diagnosis

The specific objectives are:

- Evaluate the diagnostic performance and confidence scores of UniMod.
- Measure confidence calibration using Expected Calibration Error (ECE) and the Brier Score.
- Investigate whether temperature scaling can improve confidence calibration.
- Compare model performance before and after calibration.
- Explore whether uncertain predictions can be identified for further review.

Example:

Comparing ECE before and after calibration can help determine whether the model's confidence has become better calibrated.[**?**]

1.4 Research Questions

Purpose: Identify the questions the research will answer. **Example:** This study will focus on the following questions:

1. How well calibrated are UniMod's confidence scores?
2. Can temperature scaling improve confidence calibration?
3. How does calibration affect the reliability of diagnostic predictions?
4. Can uncertainty estimation help identify predictions that need further review?

1.5 Scope and Delimitation

Purpose: Define what the study will not cover.

Content: This study will focus on confidence calibration and uncertainty estimation in UniMod using suitable medical image text data. It will evaluate diagnostic performance and calibration using metrics such as accuracy, ECE and the Brier Score. Temperature scaling and selective prediction will be considered where feasible. The study will be limited by dataset availability and access to the UniMod implementation and available

computing resources. It will not develop a complete clinical diagnostic system or claim that the results establish real world clinical safety.

Example:

If the experiments use one dataset the findings may not apply equally to every medical condition or hospital setting.

1.6 Significance of the Study

Purpose: Explain the potential value of the research.

Content: This study may help researchers better understand whether UniMod's confidence scores reflect its actual diagnostic performance. It may also provide insight into whether calibration techniques can improve the reliability of multimodal medical AI predictions. The findings could support future research into more trustworthy AI systems for healthcare.

Example:

Calibration improves the reliability of confidence scores, future studies could investigate whether the improvement also holds across other medical datasets.[**?**]

1.7 Definition of Terms (Optional)

Purpose: Clarify key concepts or technical terms for readers.

Example:

- Artificial Intelligence (AI): Technology that enables computers to perform tasks such as prediction and pattern recognition.
- Multimodal Learning: Learning from multiple types of data like images and texts.
- UniMod: The multimodal medical diagnosis framework investigated in this study.
- Confidence Calibration: Evaluating or adjusting predicted confidence so that it better reflects observed correctness.
- Uncertainty Estimation: Assessing how uncertain a model is about its predictions.
- Expected Calibration Error (ECE): A metric that measures the difference between predicted confidence and observed accuracy across groups of predictions.
- Brier Score: A metric used to evaluate the accuracy of predicted probabilities.
- Temperature Scaling: A technique that adjusts predicted probabilities to improve calibration.
- Selective Prediction: Allowing a model to abstain from making predictions on cases considered uncertain.

Chapter 2:

Literature Review

2.1 Confidence Calibration in Deep Learning

Purpose: Explain confidence calibration and why it is important for reliable predictions in deep learning especially in medical diagnosis.

Content:

Guo et al. (2017) investigated the calibration of modern neural networks and found that neural networks can produce poorly calibrated confidence estimates. They also evaluated several calibration methods and demonstrated that temperature scaling was effective across many of the settings they studied. Temperature scaling adjusts a model's predicted probabilities using a learned temperature parameter without requiring the entire model to be retrained. This method is relevant to the proposed research because it provides a practical way to investigate whether UniMod's confidence estimates can be improved. Its effectiveness on UniMod.[1]

Example:

Suppose Unimod predicts a disease with 90% confidence but only 70% of comparable predictions are correct. This suggests that the model may be overconfident. Temperature scaling can be applied to examine whether its confidence estimates become better aligned with observed accuracy.

2.2 Calibration Metrics for Evaluating Model Confidence

Purpose: Explain how calibration metrics can be used to evaluate whether a model's confidence estimates are reliable.

Content: Classification accuracy measures how often a model predicts the correct class. It does not directly measure the reliability of the probabilities produced by the model. Therefore, calibration metrics are useful for examining the relationship between predicted confidence and actual outcomes. Nixon et al. (2019) studied calibration measurement in

deep learning and discussed limitations associated with commonly used metrics. They introduced Adaptive Calibration Error (ACE). Which uses adaptive binning to assess differences between predicted confidence and observed accuracy. Their work highlights the importance of choosing appropriate metrics when evaluating calibration. In this research, Expected Calibration Error (ECE) and the Brier Score are proposed as evaluation metrics. ECE summarizes the difference between confidence and accuracy across groups of predictions. The Brier Score measures the squared difference between predicted probabilities and the actual class outcomes. Using both metrics can provide complementary information about the quality of the model's predictions. Classification accuracy will also be reported to assess predictive performance. [2]

Example:

Suppose Unimod achieves high classification accuracy but has a relatively high ECE. This could indicate that its predictions are often correct while its confidence estimates remain poorly calibrated. Comparing ECE and Brier Score before and after temperature scaling would help assess whether calibration improves.

2.3 Uncertainty Estimation in Medical Image Classification)

Purpose: Explain why uncertainty estimation is important in medical image classification and introduce common approaches used in previous research.

Content:

Medical image classification can be challenging because images may contain noise, unclear features or patterns that differ from those in the training data. In these situations, a model may produce an incorrect prediction or an unreliable confidence estimate. Uncertainty estimation methods aim to provide additional information about the reliability of model predictions. Kurz et al. (2022) conducted a systematic review of uncertainty estimation in medical image classification. Their review examined methods used in existing studies. Including Monte Carlo dropout and deep ensembles. The review also discussed evaluation approaches and the need for further research into how uncertainty information can support the use of AI in medical image analysis. These findings provide background for the proposed study. Although several methods can be used to estimate uncertainty. This research will focus on confidence calibration using temperature scaling and provided that the UniMod implementation supports the method. This narrower scope will allow the study to investigate confidence reliability without requiring the implementation of several different uncertainty estimation techniques. [3]

Example: A medical image may contain unclear features that make classification difficult. An uncertainty aware system can help identify such cases for further review rather than treating every prediction as equally reliable.

2.4 Types of Uncertainty in Medical Imaging

Purpose: Explain the main types of uncertainty in deep learning and clarify how confidence calibration relates to uncertainty estimation.

Content: Uncertainty in deep learning is commonly discussed in terms of aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty arises from ambiguity or noise in the input data. While epistemic uncertainty is associated with limitations in the model's learned knowledge for example a low quality medical image may introduce data related uncertainty. Where as an image from an unfamiliar population may expose limitations in the model's training experience. Zou et al. (2023) reviewed uncertainty estimation methods and their applications in medical imaging. Their work discusses different forms of uncertainty methods for estimating them and challenges associated with interpreting uncertainty in medical applications. This literature helps explain why model reliability cannot always be understood through a single confidence value. In the proposed research, confidence calibration will be the primary focus. Calibration evaluates whether predicted probabilities correspond to observed correctness. Where as broader uncertainty estimation may address additional sources of uncertainty. These concepts are related but they are not identical. [4]

Example: A model may assign a high probability to a diagnosis even when the input image is unclear. Calibration evaluation can help determine whether confidence values correspond to observed accuracy across predictions. Although it does not by itself identify every source of uncertainty in an individual case.

2.5 Uncertainty Quantification for Trustworthy Clinical AI

Purpose: Explain the importance of uncertainty quantification for trustworthy medical AI and connect it to the proposed evaluation of UniMod.

Content: Lambert et al. (2024) reviewed uncertainty quantification in deep learning models for medical image analysis. Their work discusses approaches to uncertainty quantification, evaluation challenges and considerations for developing more trustworthy clinical AI systems. This research provides a broader clinical context for studies that evaluate not only the predictions of an AI model but also the reliability of its outputs. For UniMod, confidence calibration could provide useful evidence about whether the model's predicted probabilities align with its observed performance. However improved calibration would not automatically demonstrate clinical safety or readiness for deployment. The findings must be interpreted within the limitations of the available model, dataset and experimental design. [5]

Example:

If UniMod assigns high confidence to predictions that are frequently incorrect. Clinicians can be misled by its output. Evaluating calibration may reveal this problem and help determine whether a calibration method improves the reliability of its confidence estimates.

2.6 Research Gap

Purpose: To identify the research question that emerges from the literature and explain how the proposed study will address it.

Content: The five reviewed papers address different but related aspects of reliable AI prediction. Guo et al. (2017) investigate confidence calibration and temperature scaling. Nixon et al. (2019) examine calibration metrics and their limitations. Kurz et al. (2022) review uncertainty estimation in medical image classification. While Zou et al. (2023) discuss uncertainty estimation in medical imaging more broadly. Lambert et al. (2024) consider uncertainty quantification in the context of trustworthy clinical AI. Together, these studies provide a theoretical and methodological foundation for investigating confidence reliability in medical AI. However, the five papers alone do not establish whether temperature scaling improves the calibration of UniMod specifically. This question requires examination of the existing UniMod implementation and an experimental evaluation using suitable data. The proposed study will compare UniMod's baseline calibration with its calibration after temperature scaling. If the model's outputs and available validation data permit this procedure. ECE and Brier Score will be used to evaluate calibration alongside classification accuracy. The results will indicate whether temperature scaling improves calibration under the experimental conditions. [1] [2] [3] [4] [5]

Example:

The study could calculate ECE and Brier Score for the baseline UniMod model. It can apply temperature scaling using a validation set and calculate the metrics again on an appropriate evaluation set. A comparison would show whether calibration improved in that experiment.

2.7 Chapter Summary

Purpose: To summarize the key ideas from the literature review and establish a transition to the research methodology.

Content: This chapter reviewed five studies related to confidence calibration, calibration metrics, uncertainty estimation in medical image classification, types of uncertainty and uncertainty quantification for clinical AI. The literature highlights the importance of evaluating confidence estimates alongside classification performance. It also provides a basis for investigating temperature scaling and using ECE and Brier Score to evaluate calibration.

Chapter 3:

Methodology

3.1 Research Design

Purpose: To explain the overall research approach and how the study will investigate confidence calibration in UniMod.

Content: This study will use an experimental research design to examine whether temperature scaling can improve the confidence calibration of UniMod for multimodal medical diagnosis. The study will compare the original model with a calibrated version using the same held out test data for both evaluations. The evaluation will focus on three metrics: classification accuracy, Expected Calibration Error (ECE) and Brier Score. Accuracy will measure the proportion of correct predictions. While ECE and Brier Score will help assess the reliability of the predicted probabilities. This approach follows the broader principle that the quality of a model's confidence estimates should be evaluated alongside its predictive performance. The study will not assume that UniMod is poorly calibrated or that temperature scaling will necessarily improve its performance. Instead, these questions will be investigated through experiments. [1] [2]

Example:

The original UniMod model will first be evaluated to establish baseline results. Temperature scaling will then be fitted using validation data and the calibrated predictions will be evaluated on the same test examples to determine whether the selected metrics improve.

3.2 Proposed Research Pipeline

Purpose: To present the sequence of activities that will be followed during the research.

Content: The proposed pipeline consists of reviewing the UniMod implementation. Selecting an accessible dataset, preparing the data, evaluating the baseline model, fitting temperature scaling, calculating evaluation metrics and comparing the results. The validation set will be used to determine the temperature parameter. The test set will remain

separate from this calibration process and will be used only for the final evaluation. This separation will help prevent information from the test labels from influencing the calibration procedure.

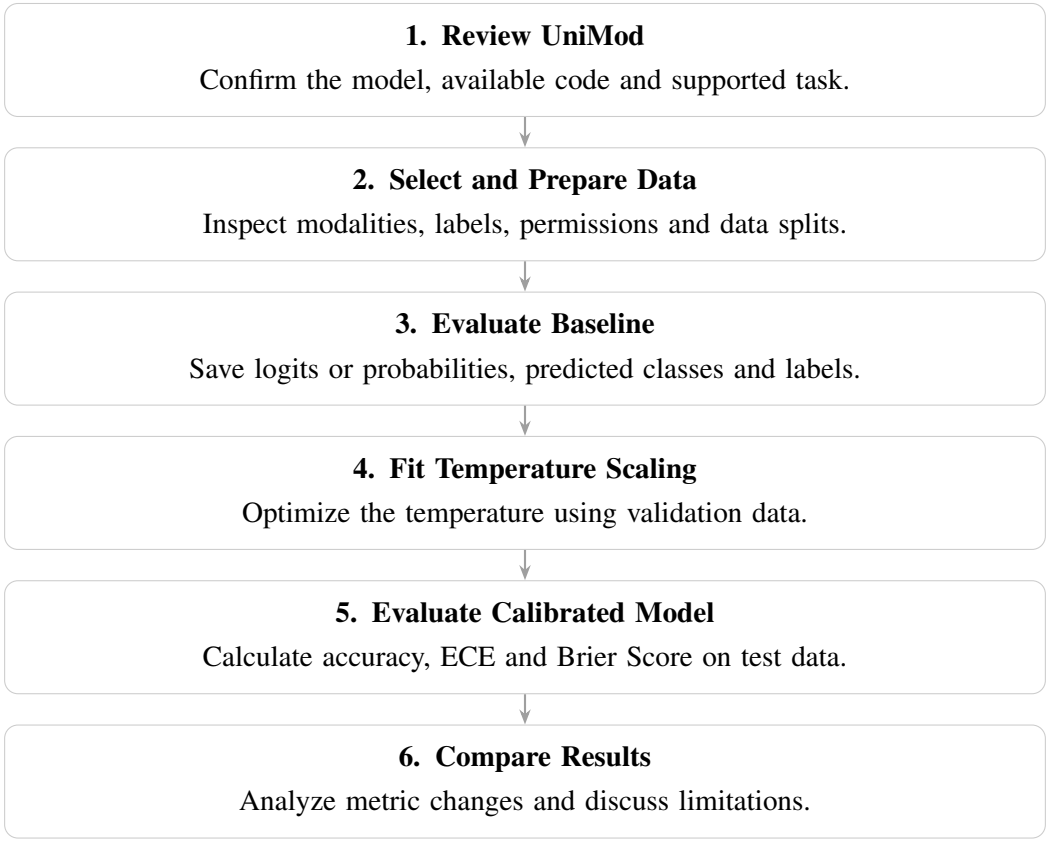


Figure 3.1: Proposed research pipeline for evaluating confidence calibration in UniMod.

Example:

If the baseline model produces overconfident predictions. The calibration procedure will test whether adjusting its logits reduces the difference between predicted confidence and observed accuracy. [1] [2]

3.3 Dataset Selection and Preparation

Purpose: To explain which data will be needed and how it will be prepared for the experiments.

Content: The study will use a dataset that is compatible with the available UniMod implementation. The final dataset and diagnostic task will be selected after reviewing the original UniMod paper, checking the repository and confirming access to the required model weights and data. Depending on the supported task the dataset may contain medical images, associated clinical text, diagnostic labels or other modalities. The data will be inspected for missing values, label consistency, class distribution and preprocessing

requirements. Where the original setup permits the data will be separated into training, validation and test sets. The validation set will be used to fit the temperature parameter. While the test set will be reserved for the final comparison. Existing recommended dataset splits will be followed when available. If multiple records belong to the same patient, patient level separation will be used where appropriate to reduce the risk of data leakage. Preprocessing, including image resizing, normalization and text tokenization will follow the original UniMod implementation as closely as possible.

Example:

If the selected task uses medical images paired with diagnostic text the inputs will be prepared using the model's existing preprocessing procedure. Each example will be associated with the corresponding ground truth label so that the predictions can be evaluated. [3] [4] [5]

3.4 Baseline UniMod Evaluation

Purpose: To establish the model's original performance before applying temperature scaling.

Content: The baseline experiment will be conducted using the available UniMod model and its supported diagnostic task. A pretrained checkpoint will be used if it is accessible and suitable for the selected dataset. The model will process the required input modalities and produce predictions for the relevant diagnostic classes. For each test example, the predicted class probability values or logits and ground truth label will be saved. These outputs will be used to calculate accuracy, ECE and Brier Score. The baseline results will serve as the reference point for the calibrated model. Saving the probability values is important because classification labels alone cannot show how confident the model was in its predictions. This distinction is central to confidence calibration as discussed by Guo et al. (2017).

Example: For one medical case, UniMod may predict a particular class with a probability of 0.85. The predicted class and probability will be recorded alongside the true label. Repeating this process for the test set will provide the data needed to evaluate the model's baseline performance. [1]

3.5 Temperature Scaling

Purpose: To describe the calibration method that will be used to adjust UniMod's predicted probabilities.

Content: Temperature scaling is a post processing method introduced by Guo et al. (2017). It adjusts a model's logits using a single positive parameter called the temperature. Before converting the logits into probabilities. The method does not require the model's original weights to be retrained.

For a multiclass classification problem, the calibrated probability of class i is calculated as:

$$p_i(T) = \frac{\exp(z_i/T)}{\sum_{j=1}^{K} \exp(z_j/T)} \tag{3.1}$$

where z_i represents the logit for class i, K is the total number of classes, and T is the temperature parameter.

The temperature will be optimized on the validation set by minimizing negative log likelihood. Once the temperature has been selected it will be applied to the test logits. The test labels will not be used to choose the temperature. A temperature greater than 1 generally makes the predicted probability distribution less concentrated. While a temperature below 1 makes it more concentrated. The effect on UniMod's calibration will be determined from the experimental results rather than assumed in advance.

This method can be applied only if the available UniMod implementation exposes suitable logits or supports obtaining them. If only class labels are available, additional access to the model outputs will be required.

Example:

Suppose UniMod assigns a probability of 0.95 to one diagnostic class. If validation results indicate that its predictions are generally overconfident the learned temperature may reduce that probability. The calibrated probabilities will then be evaluated on the test set to determine whether their reliability improves. [1]

3.6 Evaluation Metrics

Purpose: To define the measures that will be used to compare the baseline and calibrated predictions.

Content: Three metrics will be used in the proposed evaluation: classification accuracy, Expected Calibration Error (ECE) and Brier Score. Accuracy will measure predictive performance while ECE and Brier Score will assess the quality of the predicted probabilities from complementary perspectives.

Nixon et al. (2019) examined calibration measurement and highlighted limitations in commonly used calibration metrics. Their work motivates careful selection and interpretation of the measures used in this study.

Classification Accuracy

Accuracy measures the proportion of test examples for which the predicted class matches the true class:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} 1(\hat{y}_i = y_i) \tag{3.2}$$

Here, N is the number of test examples, $\hat{y}_i$ is the predicted class, and y_i is the ground truth label.

Example: If the model correctly classifies 85 out of 100 test examples, its accuracy is 85%.

Expected Calibration Error (ECE)

ECE summarizes the difference between confidence and observed accuracy across groups of predictions. One commonly used definition is:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\text{acc}(S_b) - \text{conf}(S_b)\right| \tag{3.3}$$

where B is the number of confidence bins, S_b is the set of predictions in bin b, and $\text{acc}(S_b)$ and $\text{conf}(S_b)$ are the average accuracy and average confidence within that bin.

A lower ECE generally indicates better calibration under the selected binning procedure. The binning method and number of bins will be documented and kept consistent between the baseline and calibrated evaluations.

Example: If predictions in a confidence bin have an average confidence of 90% but an observed accuracy of 70%, the difference contributes to the ECE. [2]

Brier Score

The Brier Score measures the squared difference between predicted probabilities and actual class outcomes. For a multiclass task, it can be calculated as:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}(p_{ik} - y_{ik})^2 \tag{3.4}$$

Here, p_{ik} is the predicted probability for class k in example i, and y_{ik} equals 1 when class k is the true class and 0 otherwise.

A lower Brier Score indicates better probabilistic predictions under this definition. However, it reflects both calibration and other aspects of probabilistic prediction quality, so it will be interpreted alongside ECE and accuracy.

Example: A model that assigns a very high probability to the wrong class will receive a larger probability error than a model that assigns a lower probability to that incorrect class. [2]

3.7 Experimental Procedure

Purpose: To explain the order in which the experiments will be carried out.

Content: The experimental procedure will begin with checking the UniMod implementation and identifying a suitable dataset and diagnostic task. The data will then be prepared according to the model's original requirements and the available model checkpoint will be used to generate baseline predictions.

The baseline predictions will be evaluated using accuracy, ECE and Brier Score. Next temperature scaling will be fitted using the validation data. The learned temperature will be applied to the test logits and the same metrics will be calculated for the calibrated predictions.

The results will be compared using the same test examples and metric definitions. Any changes to the original model, preprocessing procedure or experimental setup will be recorded. This procedure follows the use of a separate validation set for calibration described by Guo et al. (2017).

Example:

The experiment may produce an ECE of 0.12 for baseline UniMod and 0.07 after temperature scaling. If those values are obtained from the actual test evaluation the reduction would indicate improved calibration under the selected ECE method. The values here are illustrative only and must not be reported as research findings. [1] [2]

3.8 Result Analysis

Purpose: To explain how the results will be compared and how conclusions will be drawn from the experiments.

Content: The results will be presented in tables and graphs comparing the baseline and temperature scaled versions of UniMod. The main focus will be on whether ECE and Brier Score decrease after calibration and whether classification accuracy remains comparable.

A reliability diagram may be included to show the relationship between predicted confidence and observed accuracy. A well calibrated model should have confidence values that more closely correspond to observed accuracy across the plotted groups. The results will be interpreted cautiously. A reduction in ECE does not necessarily mean that every prediction has become more reliable and a lower Brier Score does not isolate calibration from all other aspects of probabilistic prediction quality. These limitations will be considered when discussing the findings.

Example:

If the calibrated model has a lower ECE and Brier Score while maintaining similar accuracy, the findings would support the conclusion that temperature scaling improved cali-

Metric	Baseline UniMod	Calibrated UniMod
Accuracy	To be measured	To be measured
ECE	To be measured	To be measured
Brier Score	To be measured	To be measured

Table 3.1: Planned comparison of baseline and calibrated predictions.

bration under the tested conditions.

3.9 Tools and Implementation

Purpose: To identify the software and computational resources needed to conduct the experiments.

Content: The implementation will use the tools supported by the official UniMod repository. Python may be used if the implementation is Python based. Depending on the repository, PyTorch may be required for model inference while NumPy and pandas may be used to organize prediction outputs. Matplotlib may be used to produce reliability diagrams and other result visualizations.

The final choice of tools will depend on the available source code, model checkpoint and computing resources. Software versions, data splits, preprocessing settings and the selected calibration parameters will be documented so that the experimental procedure can be reproduced.

Example:

A Python script could load the model checkpoint, save the predicted logits and labels. Fit the temperature parameter on the validation set and calculate the selected evaluation metrics.

3.10 Ethical Considerations

Purpose: To explain how the study will handle medical data responsibly and report its findings appropriately.

Content: The study will follow the access conditions and usage restrictions of the selected dataset. If the data contain patient information, relevant privacy protections and research requirements will be followed. No attempt will be made to identify individual patients. The findings will be presented as an experimental evaluation of confidence calibration. Any improvement in calibration will be interpreted as evidence about the measured probabilities under the tested conditions, rather than as proof that UniMod is safe for clinical deployment.

Example:

If the dataset requires approval for access or prohibits redistribution the researcher will follow those conditions. Results may be reported in aggregate without exposing restricted patient information.

3.11 Methodological Limitations

Purpose: To identify factors that could restrict the study or affect the interpretation of its results.

Content: The study may be limited by access to the UniMod source code, pretrained model weights or suitable datasets. Temperature scaling also requires access to appropriate logits and validation labels. If these outputs are unavailable the proposed calibration procedure will need to be revised.

Another limitation is that calibration metrics provide different and incomplete views of prediction reliability. Nixon et al. (2019) discuss shortcomings in commonly used calibration measurements while Kurz et al. (2022) review the range of uncertainty estimation approaches used in medical image classification. These studies highlight why calibration results should be interpreted within the context of the selected method and task. The findings will apply primarily to the particular model, dataset and evaluation setup used in the experiments. They may not generalize to other diagnostic tasks or patient populations. Furthermore, temperature scaling adjusts probabilities but does not independently identify every source of uncertainty or correct all model errors

Example:

If temperature scaling improves ECE on one dataset. It does not automatically follow that the calibrated model will achieve the same improvement on a different medical dataset. [2] [3] [5]

3.12 Chapter Summary

Purpose: To summarize the proposed methodology and prepare the reader for the next chapter.

Content: This chapter presented the proposed experimental approach for evaluating confidence calibration in UniMod. The study will establish baseline performance, apply temperature scaling using validation data and compare the baseline and calibrated predictions using accuracy, ECE and Brier Score. The results will be evaluated on held out test data and interpreted in light of the limitations of the selected metrics and experimental setup.

Example:

After completing the experiments the results chapter will report the measured metrics. Include relevant plots and discuss whether the findings support an improvement in confidence calibration.

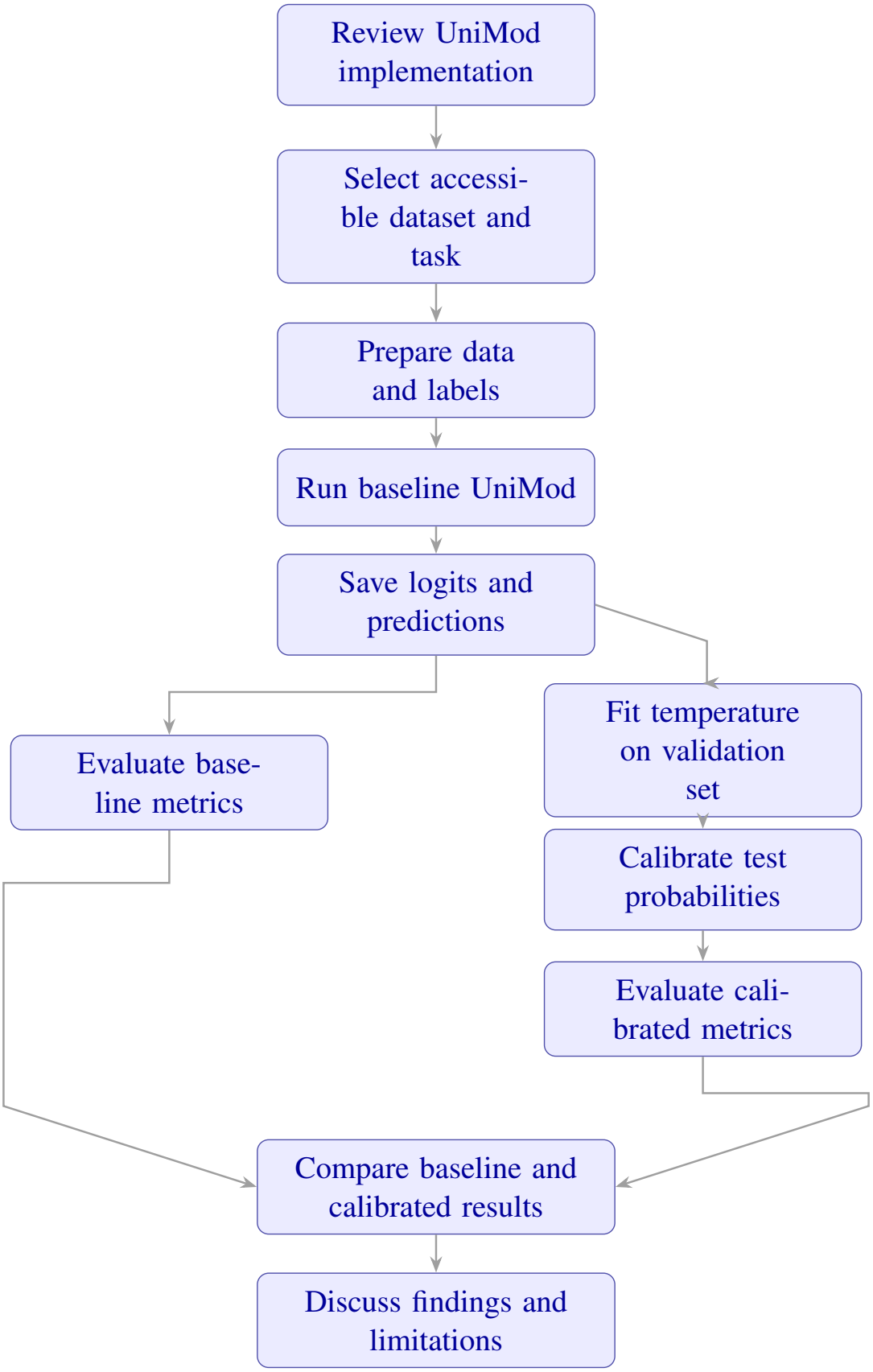


Figure 3.2: Experimental workflow for confidence calibration in UniMod.

3.13 Summary of Citation Placement

Paper	Where to cite it in Chapter 3	Main purpose
Guo et al. (2017)	Sections 3.1, 3.4, 3.5, 3.7, 3.8	Confidence calibration, temperature scaling and validation based calibration.
Nixon et al. (2019)	Sections 3.1, 3.6, 3.7, 3.8, 3.11	Calibration metrics and their limitations.
Kurz et al. (2022)	Sections 3.3 and 3.11	Uncertainty estimation methods in medical image classification.
Zou et al. (2023)	Section 3.11, or wherever uncertainty types are discussed	Aleatoric and epistemic uncertainty in medical imaging.
Lambert et al. (2024)	Sections 3.8, 3.10, and 3.11	Uncertainty quantification and challenges in trustworthy clinical AI.

Table 3.2: Summary of citation placement in Chapter 3.

Chapter 4:

Results and Analysis

4.1 Introduction

Purpose: To introduce the chapter and explain what results will be presented and analyzed.

Content: This chapter presents the results of the experiments conducted to evaluate confidence calibration in UniMod for multimodal medical diagnosis. The main objective is to investigate whether temperature scaling can improve the reliability of the model's predicted confidence without significantly affecting its diagnostic performance.

The baseline UniMod model will first be evaluated using the selected test dataset. Its performance will then be compared with the performance of the temperature scaled model. Accuracy, Expected Calibration Error (ECE) and Brier Score will be used to assess the results. Reliability diagrams may also be used to visualize the relationship between predicted confidence and actual prediction accuracy.

Example:

The experiments compare the baseline UniMod model with a temperature scaled version of the same model. Both models are evaluated using the same test dataset to ensure a fair comparison. The results are analyzed in terms of predictive performance and confidence reliability. Particular attention is given to whether temperature scaling reduces calibration error while maintaining similar classification accuracy.

4.2 Experimental Setup

Purpose: To describe the experimental conditions under which the results are obtained.

Content: The experiments will use the UniMod model, selected medical diagnosis dataset and the evaluation procedure described in Chapter 3. The dataset will be divided into training, validation and test sets according to the dataset's official protocol or the chosen experimental design.

The baseline model will be evaluated without additional calibration. Temperature scaling will then be fitted using the validation set. The learned temperature value will be applied to the model's test set logits to obtain calibrated confidence scores. The test set will not be used to select the temperature value.

Both versions of the model will be evaluated using the same test examples and evaluation metrics. This approach will help ensure that any observed differences in calibration are associated with the calibration procedure rather than changes in the evaluation data. The implementation details including the dataset name, model checkpoint, data split, software libraries and hardware should be reported once they are confirmed.

4.3 Baseline UniMod Performance

Purpose: To establish the performance of the original UniMod model before applying confidence calibration.

Content: The baseline evaluation provides a reference point for measuring the effect of temperature scaling. The model's predictions will be assessed using accuracy, ECE and Brier Score.

Accuracy measures the proportion of test examples classified correctly. ECE measures the difference between model confidence and observed accuracy across confidence groups. The Brier Score measures the difference between predicted class probabilities and the true class labels. These metrics provide complementary information. A model may achieve high accuracy while producing confidence scores that do not accurately reflect its likelihood of being correct. Therefore, baseline performance should be interpreted in terms of both classification quality and confidence reliability.

Example:

Metric	Baseline result
Accuracy (%)	To be measured
ECE	To be measured
Brier Score	To be measured

Table 4.1: Baseline UniMod results on the test dataset.

4.4 Temperature Scaling Results

Purpose: To examine how temperature scaling changes UniMod's confidence estimates.

Content:

Temperature scaling adjusts the model's predicted class probabilities by dividing the logits by a positive temperature parameter before applying the softmax function. The model's learned weights remain unchanged. Only the confidence distribution is adjusted. When the learned temperature is greater than 1 the predicted probability distribution generally

becomes less concentrated, reducing overconfident predictions. When the temperature is between 0 and 1 the distribution generally becomes more concentrated. The effect of temperature scaling on the model must be assessed using the actual experimental results. The learned temperature value and the calibration metrics before and after scaling will be reported. A reduction in ECE or Brier Score would indicate an improvement in the corresponding measure. While accuracy should also be checked to determine whether classification performance remains stable.

Example:

Item	Result
Learned temperature, T	To be measured
Baseline ECE	To be measured
Calibrated ECE	To be measured
Baseline Brier Score	To be measured
Calibrated Brier Score	To be measured

Table 4.2: Temperature scaling results.

4.5 Comparison Between Baseline and Calibrated UniMod

Purpose: To compare the two model configurations and determine whether calibration improves confidence reliability while preserving diagnostic performance.

Content:

The baseline and temperature scaled versions of UniMod will be compared using the same test set. Accuracy will be used to assess classification performance. While ECE and Brier Score will be used to assess probability quality and calibration related performance. The difference in accuracy will show whether the calibration procedure changes the model's classification performance. The changes in ECE and Brier Score will indicate whether the predicted probabilities improve according to these metrics. Since temperature scaling is applied to the logits without retraining the underlying model, predicted class labels will generally remain unchanged when the temperature is positive and the same logits are used.

Example:

Metric	Baseline UniMod	Calibrated UniMod	Interpretation
Accuracy (%)	To be measured	To be measured	Classification performance
ECE	To be measured	To be measured	Lower is generally better
Brier Score	To be measured	To be measured	Lower is generally better

Table 4.3: Comparison of baseline and calibrated UniMod.

4.6 Expected Calibration Error Analysis

Purpose: To examine how closely UniMod's predicted confidence agrees with its observed accuracy.

Content:

Expected Calibration Error summarizes the difference between confidence and accuracy across groups of predictions. Usually formed by dividing predictions into confidence bins. A lower ECE generally indicates closer agreement between predicted confidence and observed accuracy for the selected binning scheme. ECE will be calculated for both the baseline and calibrated models using the same binning strategy. The number of bins and the binning method should be reported because ECE values can change depending on these choices. ECE also has limitations, particularly when some bins contain relatively few predictions. The ECE results will be interpreted alongside accuracy and the reliability diagram rather than being treated as a complete measure of uncertainty.

Example: Suppose the baseline ECE is measured as 0.12 and the calibrated ECE as 0.05. These are illustrative numbers only, not your experimental results. Such a decrease would suggest better agreement between confidence and accuracy under the selected ECE calculation.

4.7 Brier Score Analysis

Purpose: To assess whether temperature scaling improves the quality of the predicted class probabilities.

Content:

The Brier Score measures the squared difference between predicted probabilities and the corresponding true class labels. For multiclass classification the score is calculated across all classes and averaged over the evaluated examples. Lower values generally indicate better probabilistic predictions. The baseline and calibrated Brier Scores will be compared

to determine whether temperature scaling improves the overall probability predictions. However the Brier Score reflects more than calibration alone. It is also influenced by how well the model distinguishes between correct and incorrect outcomes. Therefore changes in the Brier Score should be interpreted alongside ECE and accuracy.

Example: Suppose the baseline Brier Score is 0.24 and the calibrated score is 0.18. These values are hypothetical and must not be presented as actual findings. If your real results show a decrease you can state that the calibrated model produced better probability predictions according to the Brier Score on the evaluated test set.

4.8 Reliability Diagram Analysis

Purpose: To visualize the relationship between UniMod's confidence scores and the observed accuracy of its predictions.

Content:

A reliability diagram groups predictions by confidence and compares the average confidence in each group with the observed accuracy. A perfectly calibrated model would have average confidence close to observed accuracy across the groups. Reliability diagrams will be generated for the baseline and temperature scaled models using the same confidence bins. The diagrams will be compared to identify whether temperature scaling moves the model's confidence estimates closer to the ideal calibration line. The diagrams will be interpreted together with ECE because visual patterns can reveal differences that a single summary score may not fully describe. Results should also be interpreted carefully when a bin contains only a small number of predictions.

Example:

4.8.1 Illustrative reliability diagram interpretation

Concept:

[1] [2]

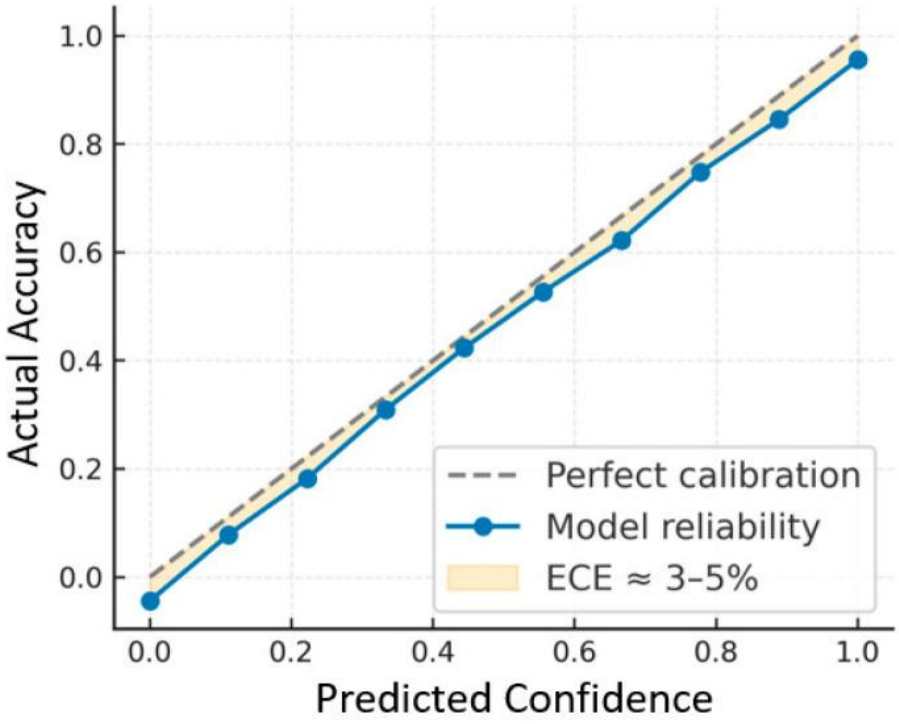


Figure 4.1: Illustrative reliability diagram showing predicted confidence versus actual accuracy.

4.9 Discussion of the Main Findings

Purpose: To explain what the combined experimental results mean in relation to the research objectives.

Content:

The findings will be discussed in terms of three main questions:

1. Does baseline UniMod produce confidence scores that are well calibrated?
2. Does temperature scaling improve calibration according to ECE, Brier Score and the reliability diagram?
3. Does the calibration procedure preserve the model's classification accuracy?

If calibration improves while accuracy remains stable the findings would support the use of temperature scaling as a practical post processing method for the evaluated UniMod configuration. If the metrics provide mixed results the discussion should explain those differences rather than selecting only the most favorable metric.

Example: If the experimental results show that temperature scaling reduces ECE and Brier Score while leaving accuracy unchanged the discussion could explain that the method improved the reliability of the model's probability estimates without altering its classification decisions on the test set. This would be consistent with the purpose of temperature

scaling described by Guo et al. (2017). If the results do not show improvement, the discussion should state that temperature scaling did not provide a clear benefit under the chosen experimental conditions. Possible limitations, including dataset size, validation set representativeness and access to the model's logits can then be considered. [1]

4.10 Limitations of the Results

Purpose: To acknowledge the limitations that affect the interpretation and generalizability of the experimental findings.

Content:

The results may be affected by the size and composition of the selected dataset. The quality of its labels and the representativeness of the validation set used to fit the temperature parameter. ECE can also depend on the binning strategy while the Brier Score reflects both calibration and other aspects of probability prediction. If only one UniMod checkpoint or one dataset is evaluated the findings should not be generalized to all UniMod configurations or medical diagnosis tasks. Likewise improvements on a test dataset do not establish that the model will remain calibrated on data from different hospitals, patient populations, devices or acquisition settings.

Example: The experiment evaluates temperature scaling on one selected UniMod configuration and one test dataset. Therefore the results indicate the calibration performance observed under those conditions and may not represent performance on other medical datasets. Further evaluation across additional datasets and external clinical settings would be needed to assess generalizability.

4.11 Chapter Summary

Purpose: To summarize the chapter's main results and connect them to the next chapter.

Content:

This chapter presents the evaluation framework for comparing baseline UniMod with a temperature scaled version. The analysis covers classification accuracy, Expected Calibration Error, Brier Score and reliability diagrams. Together these measures provide a broader assessment of predictive performance and confidence reliability. The final summary will state whether temperature scaling improved calibration in the experiments, whether accuracy remained stable and what limitations affect the interpretation of the findings. These conclusions will be based only on the measured results.

Chapter 5:

Conclusion

5.1 Conclusion

Purpose: To summarize the main objective of the thesis and to explain the expected contribution of the research and highlight why confidence calibration is important for multimodal medical diagnosis using UniMod.

Content:

This research focuses on uncertainty estimation and confidence calibration in UniMod for multimodal medical diagnosis. Although deep learning models can provide accurate predictions, their confidence scores do not always reflect how reliable those predictions are. In medical applications, this is an important concern because an incorrect prediction accompanied by high confidence may lead users to place more trust in the model than the evidence justifies. To investigate this issue the study proposes evaluating the original UniMod model and comparing its performance with a temperature scaled version. The evaluation uses accuracy, Expected Calibration Error (ECE) and Brier Score to examine both classification performance and the reliability of predicted probabilities. Reliability diagrams can provide additional visual evidence of how closely the model's confidence aligns with its observed accuracy. The main contribution of this research is to investigate whether a relatively simple calibration technique can improve the reliability of UniMod's confidence estimates without substantially affecting its diagnostic performance. The study also highlights the importance of evaluating confidence separately from prediction accuracy when developing AI systems for medical applications.

The final conclusion will depend on the experimental findings. If temperature scaling improves the calibration metrics while maintaining similar accuracy it would suggest that the technique is useful for the evaluated UniMod configuration. If the improvement is limited or inconsistent, that outcome will also be valuable because it will show that calibration performance depends on the model and evaluation conditions.

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